

1. anandpratapsingh@lt-anandp:~\$ mkdir -p Workspace/Projects/2025/Logs
list directory: cd Workspace
ls -l
2. anandpratapsingh@lt-anandp:~/Workspace/Projects/2025\$ touch report.txt
anandpratapsingh@lt-anandp:~/Workspace/Projects/2025\$ touch data.csv
anandpratapsingh@lt-anandp:~/Workspace/Projects/2025\$ touch notes.md

anandpratapsingh@lt-anandp:~\$ mv Workspace/Projects/2025/notes.md
Workspace/Projects/2025/Logs/
3. anandpratapsingh@lt-anandp:~\$ find Workspace/Projects/ -type f -name "*.txt" | while read file; do
new_name="\${file%.txt}_backup.txt"
mv "\$file" "\$new_name"
done
4. anandpratapsingh@lt-anandp:~\$ find Workspace/ -type f -name "*.txt" | while read file; do
cp "\$file" Backup/
done
5. anandpratapsingh@lt-anandp:~\$ for i in {1..10}; do
touch Workspace/file\$i.txt
done

i=1
for file in Workspace/file*.txt; do
if ((i % 2 == 0)); then
rm "\$file"
fi
((i++))
done
6. anandpratapsingh@lt-anandp:~\$ find Workspace/Projects/2025/Logs/ -type f -mtime +7 | while read
file; do
mv "\$file" Backup/
done
7. anandpratapsingh@lt-anandp:~\$ tar -czvf workspace_backup.tar.gz Worksapce/
8. anandpratapsingh@lt-anandp:~\$ mkdir -p Restored
anandpratapsingh@lt-anandp:~\$ tar -xzf workspace_backup.tar.gz -C Restored/
9. anandpratapsingh@lt-anandp:~\$ find Workspace/ -type d -empty
anandpratapsingh@lt-anandp:~\$ find Workspace/ -type d -empty -delete
10. anandpratapsingh@lt-anandp:~\$ du -h Workspace/
11. anandpratapsingh@lt-anandp:~\$ mkdir Confidential
anandpratapsingh@lt-anandp:~\$ chmod 700 Confidential

12. anandpratapsingh@lt-anandp:~\$ sudo groupadd teamwork
anandpratapsingh@lt-anandp:~\$ touch confidential.txt
anandpratapsingh@lt-anandp:~\$ sudo chgrp teamwork confidential.txt
anandpratapsingh@lt-anandp:~\$ chmod 060 confidential.txt
13. anandpratapsingh@lt-anandp:~\$ touch executable.txt
anandpratapsingh@lt-anandp:~\$ chmod +x executable.txt
anandpratapsingh@lt-anandp:~\$ sudo mv executable.txt /usr/local/bin/
14. anandpratapsingh@lt-anandp:~\$ touch myScript.sh
anandpratapsingh@lt-anandp:~\$ chmod 700 myScript.sh
15. anandpratapsingh@lt-anandp:~\$ sudo groupadd myGroup
anandpratapsingh@lt-anandp:~\$ mkdir secure_dir
anandpratapsingh@lt-anandp:~\$ sudo chgrp myGroup secure_dir
anandpratapsingh@lt-anandp:~\$ chmod 770 secure_dir
16. anandpratapsingh@lt-anandp:~\$ sudo chown root:root Workspace/Projects/2025/Logs/*.log
17. anandpratapsingh@lt-anandp:~\$ chmod a+x Workspace/*.sh
18. anandpratapsingh@lt-anandp:~\$ find . -type f -perm 0777 -exec chmod 750 {} +
19. anandpratapsingh@lt-anandp:~\$ find Workspace/ -type f -size +100M -print0 | tar --null -czvf large_files_archive.tar.gz --files-from=-
20. anandpratapsingh@lt-anandp:~\$ grep -r "error" Workspace/ > error_summary.txt
21. anandpratapsingh@lt-anandp:~\$ grep -o "critical" Logs/system.log | wc -l
22. anandpratapsingh@lt-anandp:~\$ find . -type f -name "*.txt" -mtime -2
23. anandpratapsingh@lt-anandp:~\$ find Workspace/ -type f -user newuser
24. anandpratapsingh@lt-anandp:~\$ sudo lsof
25. anandpratapsingh@lt-anandp:~\$ sudo useradd -m -G developers devuser
26. anandpratapsingh@lt-anandp:~\$ sudo passwd devuser
anandpratapsingh@lt-anandp:~\$ sudo usermod -L devuser
27. anandpratapsingh@lt-anandp:~\$ sudo mkdir /shared_data
anandpratapsingh@lt-anandp:~\$ sudo chown root:developers /shared_data
anandpratapsingh@lt-anandp:~\$ sudo chmod 770 /shared_data
28. anandpratapsingh@lt-anandp:~\$ cat /etc/passwd
getent group developers

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29. anandpratapsingh@lt-anandp:~$ sudo userdel -r testuser
30. anandpratapsingh@lt-anandp:~$ sudo chsh -s /bin/bash devuser

31. anandpratapsingh@lt-anandp:~$ ps -eo pid,ppid,cmd,%cpu,%mem --sort=-%cpu | head -n 6 >
top_cpu_processes.txt

32. anandpratapsingh@lt-anandp:~$ sudo pkill -u testuser

33. anandpratapsingh@lt-anandp:~$ ps aux | grep apache
anandpratapsingh@lt-anandp:~$ sudo systemctl restart apache2

34. anandpratapsingh@lt-anandp:~$ top -b -d 1 -n 10

35. anandpratapsingh@lt-anandp:~$ pkill firefox

36. anandpratapsingh@lt-anandp:~$ ps -eo pid,ppid,cmd,%mem,%cpu --sort=-%mem | awk '$4 > 50'

37. anandpratapsingh@lt-anandp:~$ scp Workspace/report.txt
username@remote\_host:/destination/path/

38. anandpratapsingh@lt-anandp:~$ sudo ss -tuln

39.

40. anandpratapsingh@lt-anandp:~$ wget http://example.com/file.zip

41. anandpratapsingh@lt-anandp:~$ ping -c 4 8.8.8.8

42. anandpratapsingh@lt-anandp:~$ ip addr show
anandpratapsingh@lt-anandp:~$ ip route | grep default

43. #!/bin/bash

SOURCE="$HOME/Workspace"
DEST="$HOME/Backup"
DATE=$(date +%F)
ARCHIVE_NAME="workspace_backup_$DATE.tar.gz"

# Create Backup directory if it doesn't exist
mkdir -p "$DEST"

# Create compressed archive
tar -czf "$DEST/$ARCHIVE_NAME" "$SOURCE"

crontab -e

44. find /home/yourname/Workspace/Projects/2025/Logs/ -type f -name "*.log" -mtime +7 -exec rm -v
{} \;
```

45. #!/bin/bash

THRESHOLD=90

Check each mount point

df -h --output=pcent,target | tail -n +2 | while read percent mountpoint; do
usage=\$(echo \$percent | tr -d '%')

if ["\$usage" -ge "\$THRESHOLD"]; then

echo "[\$(date)] WARNING: Disk usage on \$mountpoint is \${usage}%!" >> ~/disk_alerts.log

fi

done

46. echo "uptime"

uptime

echo "cpu usage"

top -bn1 | grep "Cpu(s)"

echo "Memory Usage:"

free -h

47. nano users.txt

nano create_users.sh

USER_FILE="users.txt"

DEFAULT_PASS="Welcome123"

if [! -f "\$USER_FILE"]; then

echo "Error: \$USER_FILE not found!"

exit 1

fi

while IFS= read -r username; do

if id "\$username" &>/dev/null; then

echo "User \$username already exists. Skipping."

else

sudo useradd -m "\$username"

echo "\$username:\$DEFAULT_PASS" | sudo chpasswd

echo "Created user: \$username with default password."

fi

done < "\$USER_FILE"

chmod +x create_users.sh

./create_users.sh

48. SOURCE="\$HOME/Workspace"

DEST="\$HOME/Backup"

```
DATE=$(date +%F)
ARCHIVE_NAME="workspace_backup_$(date +%F).tar.gz"

mkdir -p "$DEST"

tar -czf "$DEST/$ARCHIVE_NAME" "$SOURCE"

cd "$DEST"
ls -t workspace_backup_*.tar.gz | tail -n +8 | xargs -r rm --

echo "Backup done. Old backups cleaned."
```

49. #!/bin/bash

```
REMOTE_USER="anandpratapsingh"
REMOTE_HOST="192.168.1.100"

# Generate SSH key if it doesn't exist
if [ ! -f "$HOME/.ssh/id_rsa.pub" ]; then
    ssh-keygen -t rsa -b 2048 -N "" -f "$HOME/.ssh/id_rsa"
    echo "SSH key generated."
fi

# Copy public key to remote server
ssh-copy-id "$REMOTE_USER@$REMOTE_HOST"

echo "SSH key installed for $REMOTE_USER@$REMOTE_HOST"
```

50. #!/bin/bash

```
TARGET_DIR="$HOME/Workspace"

find "$TARGET_DIR" -type d -empty -delete

crontab -e
```